Positivity Criteria for the Reverse Direction of Weak BSD in Rank Two:

A Lean 4 Formalization Skeleton with Rigorous Interval Arithmetic

Liu Xudong, Email: shtclxd8@163.com (Wenzhou Youming Machinery Technology Co., Ltd.)
Wang Fang, Email: wangfang@mail.njust.edu.cn (Nanjing University of Science and Technology)
Liu Gaoyuan, Email: 575807343@qq.com (AVIC UAV)
Liu Xiaochun, Email: 1937379994@qq.com (Chengdu University of Traditional Chinese Medicine)
Project repository: https://github.com/liuxudong8/Windows-RH-Lean-PC2
Draft of 2026-10-30 (for arXiv math.NT submission)

Abstract. The Birch–Swinnerton-Dyer (BSD) conjecture is one of the seven Millennium Problems of the Clay Mathematics Institute; its weak form asserts that the order of vanishing of the L-function at the central point s = 1 equals the algebraic rank. We address the rank-two case of weak BSD over the quadratic field $K = \mathbb{Q}(\sqrt{5})$ and construct a proof skeleton for the positivity criterion P*: if the analytic rank of E/K is exactly 2, then $L''(E/K, 1) > 0$. Our method has a two-layer structure. At the instance layer, rigorous interval arithmetic machine-closes the positivity of the central second derivative for the concrete curve 389a1 and for six curves of type (1,1), with no dependence on any conjecture. At the skeleton layer, the Lean 4 theorem prover formalizes the argument, with axioms stratified and the assumption closure audited round by round. Stage results include: the double immunity of P* to the refined BSD formula and to the generalized Riemann hypothesis (GRH is retired from the hypothesis family); the formalization skeletons of Corollary A (a lower bound for the height of the nearest non-central zero) and Corollary C (sign preservation under extension by the functional equation). We distinguish clearly between a skeleton and a complete proof: the genuine mathematical gap of P* converges to the finiteness of Sha in the $\mathbb{Q}$-side rank-two component (the 389a1 isogeny class), a single true conjecture of the refined BSD formula; under the K-side (1,1) decomposition the finiteness of Sha(E/K) is a theorem, obtained by rank symmetry (two-sided Kolyvagin combined with quadratic descent); the finiteness of Sha in the rank-one component is already Kolyvagin's theorem; and the unconditional sign-preservation channel routes the formula dependence around the assumption closure of P*. This work probes the applicability boundary of formal proof tools in arithmetic geometry and provides a re-verifiable narrowing demonstration of the hypothesis family in BSD research.

Keywords: weak BSD conjecture; elliptic curves; L-functions; Lean 4 formalization; interval arithmetic; quadratic fields

1 Introduction: the BSD conjecture and weak BSD

1.1 Background

Let E be an elliptic curve over a number field K and let L(E/K, s) be its Hasse–Weil L-function. By the modularity theorem [4], every elliptic curve over $\mathbb{Q}$ corresponds to a weight-2 modular form, whose L-function admits an Euler product, holomorphic continuation and a functional equation; over a general number field the analogous properties are supplied by the theory of automorphic L-functions [6]. The BSD conjecture consists of two assertions [1,9]. The first (weak BSD) is that the order of vanishing of L(E/K, s) at s = 1 equals the Mordell–Weil rank rank E(K), i.e. that the analytic and algebraic ranks coincide; the second (the refined formula) is that the leading Taylor coefficient equals $\Omega \cdot R \cdot |Sha| / |T|^2$, where Ω is the period integral, R the determinant of the Néron–Tate height matrix, Sha the Tate–Shafarevich group and T the torsion subgroup. The conjecture rests on a large body of numerical evidence and is regarded as a central problem at the junction of number theory and arithmetic geometry; the general case remains open.

1.2 Weak BSD and known results

For weak BSD, the deepest known results concern analytic rank at most 1. The Gross–Zagier formula [2] relates the height of Heegner points to the central derivative of the L-function: when the analytic rank is exactly 1, there exists a non-torsion rational point given by a Heegner point, so the algebraic rank is at least 1; Kolyvagin's Euler-system method [3] controls the Selmer group in the opposite direction and, within the class of modular curves, proves that the algebraic rank equals the analytic rank whenever the analytic rank is at most 1 (the finiteness of the Tate–Shafarevich group follows as well). The modularity theorem [4] (Wiles 1995 for the semistable case, completed by Breuil–Conrad–Diamond–Taylor 2001) shows that all elliptic curves over $\mathbb{Q}$ are modular, so the above results cover every elliptic curve over $\mathbb{Q}$. The reverse direction (algebraic rank $\Longrightarrow$ analytic rank) is still incompletely resolved: for algebraic rank 0 partial proofs exist under various p-adic hypotheses (Kato, Skinner–Urban, and others), while algebraic rank 1 $\Longrightarrow$ analytic rank 1 remains a core open problem. The reverse direction splits into a vanishing layer and an exclusion layer. The vanishing layer (ord $\geq$ rank) over $\mathbb{Q}$ follows by contrapositive from the theorems above (rank $\geq 2 \Longrightarrow$ ord ≥ 2 is a theorem); the exclusion layer (ord $\leq$ rank) must exclude higher-order zeros, equivalently the sign of the central derivative L^(rank)(1) > 0 — for the rank-two reverse direction the exclusion layer is exactly the sign assertion $L'' > 0$, which is the landing point of

the proposition P*. The case rank ≥ 2 (in particular both layers over a number field K) is entirely open. It is important to stress the asymmetry between ranks 1 and 2: the non-vanishing of the central first derivative $L'(1) \neq 0$ in rank 1 rests on the deep theorems of Kolyvagin and Gross–Zagier, whereas the non-vanishing of the central second derivative $L''(1) \neq 0$ in rank 2 is free at the level of definitions from the Taylor structure (ord = 2 means the $(s-1)^2$ coefficient is non-zero), with no Euler system required; the only point where rank 2 lags behind in full generality is the finiteness of Sha in the $\mathbb{Q}$-side rank-two component — currently even "analytic rank 2 $\Longrightarrow$ algebraic rank ≥ 2" (the rank-two case of the forward BSD direction) has no proof.

1.3 Positioning of our work

We do not attempt to prove weak BSD itself. Instead we focus on obstacle elimination for the rank-two reverse direction (algebraic rank $\Longrightarrow$ analytic rank) over the quadratic field $K = \mathbb{Q}(\sqrt{5})$: we study the proof path of the assertion "when the analytic rank is exactly 2, $L''(E/K, 1) > 0$", denoted P*. P* is the sign tool of the exclusion layer of the reverse proof — if the central second derivative is positive in the rank-two case, higher-order zeros (ord ≥ 3) are excluded and the reverse-direction gap over $\mathbb{Q}$ is narrowed; moreover "analytic rank 2" does not correspond to a degeneration of the regulator or to an ambiguity in the order of vanishing. We supply two kinds of evidence in parallel: machine-closed proofs by rigorous interval arithmetic at the instance layer, and formalization in the Lean 4 theorem prover at the skeleton layer.

2 Setup: $K = \mathbb{Q}(\sqrt{5})$ and the proposition P*

2.1 Choice of the number field

The choice of $K = \mathbb{Q}(\sqrt{5})$ is motivated by the unusually clean arithmetic structure of this field. First, K has class number 1; its ring of integers $\mathbb{Z}[(1+\sqrt{5})/2]$ is a unique factorization domain, so ideal theory degenerates to element theory and the computation of local factors and root numbers carries no class-group correction term. Second, the discriminant of K is 5, the smallest positive discriminant of a real quadratic field; the corresponding Hilbert modular forms and elliptic curve data are small in size and easy to verify one by one. Third, the unit group of K is generated by the golden ratio $\varphi = (1+\sqrt{5})/2$, and the explicit units give closed forms in height computations. These properties ensure that L-functions of elliptic curves over K are not disturbed by singularities in the numerical evaluation and interval-arithmetic

pipeline — the Γ-factors, the root number and the local factors can all be evaluated rigorously, which is precisely the field condition under which interval closure reaches machine precision (relative error of order 10^{-4}).

2.2 The proposition P*

Proposition P*: if the analytic rank of E/K is exactly 2, then $L''(E/K, 1) > 0$. The proof of P* decomposes into an analytic side and an algebraic side. The analytic side must show that $L''(E/K, 1)$ is genuinely positive — this is itself a sign assertion about a transcendental object. The algebraic side must show that this positivity "should" hold under weak BSD in rank two and does not come from any degeneration. In our work the two paths correspond respectively to the BSD-formula channel and the unconditional sign-preservation channel, and they finally merge.

2.3 P* and weak BSD

In the rank-two case of weak BSD, the refined formula gives $L''(E/K, 1)/2 = \Omega \cdot R_2 \cdot |Sha|/|T|^2$, where R_2 is the determinant of the 2×2 height matrix. The height pairing is given by Néron–Tate theory; its positive definiteness is a geometric rigidity fact (unconditional), so $R_2 > 0$; Ω and $|T|^2$ are also positive. Checking factor by factor: Ω, R_2, $|T|^2$ and the Tamagawa numbers are theorem-level positive quantities, and Sha (if finite) is the only mysterious factor — as soon as it is finite, the right-hand side is a product of positive real numbers and the sign holds automatically. The positivity of L'' is therefore entirely governed by the finiteness of Sha (part of the refined BSD formula) — under the (1,1)-type decomposition this finiteness is already supplied by rank symmetry (see 4.7), and the residual sign gap of the formula channel lies only in the (2,0)/(0,2) allocations containing a $\mathbb{Q}$-side rank-two component. P* is the "exclusion-layer tool" (sign layer) of the rank-two reverse direction of weak BSD: it reduces the "exclusion of higher-order zeros" required by the reverse direction to a decidable positivity proposition. Our working principle is to approach P* as closely as possible without presupposing the formula or the generalized Riemann hypothesis (GRH), and to quantify the hypotheses used at every step — any conjecture that can be bypassed is kept out of the closure.

Here the content of "finiteness of Sha" must be made precise. The Tate–Shafarevich group Sha(E/K) measures the failure of the Hasse principle (local–global principle): its elements are principal homogeneous spaces (torsors) C/K over E such that C has a point in every completion K_v but no global point — "phantom curves" that have local points

everywhere yet no global point. Its properties come in three layers. First, Sha is a torsion group (every element has finite order) — a theorem of descent arguments. Second, for every fixed n, the n-torsion Sha[n] is finite (Sha[n] embeds into $H^1(G_K, E[n])$, whose finiteness follows from Kummer theory and class field theory) — again a theorem. Third, Sha is finite in the strong sense — there exists N such that Sha[N] = Sha, i.e. the orders of all elements have a common bound — and this is the exact meaning of "finiteness of Sha" in the refined BSD formula; it is completely open in rank ≥ 2, with the exception of the (1,1)-type decomposition over a quadratic field K, where the finiteness of Sha(E/K) is closed by combining rank-one theorems through rank symmetry (see 4.7). For n-descent, the exact sequence $0 \to E(K)/nE(K) \to Sel^n \to Sha[n] \to 0$ shows that the n-component of Sha is directly accessible to descent computations; on the E^5 side of 37a1 we have confirmed Sha[2] = 0 by Selmer computation, but the overall finiteness requires the information of all n, which computational descent can never reach. Thus "finiteness of Sha" is not "every element is finite" (already a theorem) but "the set of local–global failures is bounded in total".

3 Methods and tools: the two-layer evidence structure

3.1 Instance layer: rigorous interval arithmetic

For a concrete curve, the central derivative of L(E/K, s) can be evaluated from a convergent series. The numerical pipeline is implemented with the interval-arithmetic module of mpmath at a working precision of 90 decimal digits: the incomplete Γ-function is evaluated by a self-written three-branch routine (series region, Stieltjes region and asymptotic region, each with a rigorous remainder bound), and the central-derivative series $F(s) = S_1 + N^{1-s}(2\pi)^{2s-2}\Gamma(s)^{-1}\cdot S_2'/\varepsilon$ falls term by term into closed intervals. For every target value the pipeline outputs a closed interval [a, b], and a machine check that $a > 0$ completes the closed proof for that instance. This constitutes a zero-conjecture positivity proof at the instance level: it depends neither on the BSD formula nor on GRH, only on analytic series and numerical analysis; every closed proof can be recomputed with the same pipeline.

3.2 Skeleton layer: Lean 4 formalization

The theorem-prover version of the argument uses Lean 4 (mathlib4, v4.34.0-rc2) and is laid down in modules: Positivity (Channel A, formula path: the refined BSD formula implies positivity), PositivityGRH (Channel B, sign-preservation path: zero-free regions and interval sign preservation), PStarOr (the two channels merge into a single theorem), ZeroGap

(Corollary A: lower bound on zero height), Probe389 and Probe5077 (numerical probes), and, new in this round, ExclusionReduction (the rank-one reverse exclusion layer), LeibnizReduction (task 1/2: leibniz_artin as a genuine second-order theorem), LPrime43 and LSeries5_43 (the 43a1 (1,0)-type closed proof), SymmetryReduction (rank-symmetry generic: rank_additivity, Kolyvagin ×2, descent registration, symbol_symmetry_11, p_star_11_sym), SymmetryClosure (task 4: the 79a1 (1,1)-type three-piece assembly), SelmerDescent (task 3: Selmer-layer descent registration, at the same level as Kolyvagin) and LeibnizThird (the n = 3 skeleton: (2,1)/(1,2)-type third-order Leibniz genuine theorems). The axioms of each module are strictly stratified: object layer (L-function, root number, Taylor coefficients, etc.), data layer (coefficients of the concrete curve), conjecture level (the refined BSD formula), and theorem-level registration (mathematically established, but registered in Lean because the library is incomplete). After each compilation we run #print axioms to audit the assumption closure of a theorem and list, one by one, the axioms on which it actually depends, so that every claim is verifiable. Four refined-BSD-formula axioms deserve special comment — bsd_formula_rank_one, bsd_formula_rank_one5, bsd_formula_rank_two, bsd_formula_rank_two5 — with their mathematical status graded as follows: on the rank-one side, the “Sha finiteness + sign” part is already Kolyvagin’s theorem mathematically (analytic rank 1 $\Longrightarrow$ Sha finite is proved), and the axiom registration is only an implementation gap of the formula chain in Lean; on the rank-two side the two axioms (bsd_formula_rank_two, bsd_formula_rank_two5) are genuine conjectures — the mathematical gap of P* converges to these two rank-two axioms.

3.3 Relation between the two layers

The instance closed proofs do not prove the theoretical proposition, and the theoretical skeleton does not depend on the instance numerics — the two layers of evidence stand independently and corroborate each other: the instance layer shows “the concrete curve is indeed so”, the skeleton layer shows “if the hypothesis family narrows to a certain set, the argument can be formalized”. This two-layer structure gives every claim a clear, verifiable carrier, and makes “hypothesis dependence” itself an auditable object: the closure audit report marks clearly which axioms are conjectures, which are theorem-level registrations, and which are pure logical axioms.

4 Stage results

4.1 Instance closed proofs

389a1 (conductor 389, the minimal-conductor rank-2 curve): $L''(1)/2 \in [0.7583683262976454, 0.7602647044343964]$, with the lower bound positive by machine closure. Six curves of type (1,1) (79a1, 89a1, 91a1, 99a1, 101a1, 106b1): the entangling product $L'(E,1)\cdot L'(E^5,1)$ is closed by strict intervals with positive lower bounds in all six cases (6/6), at a relative error of order 10^{-7}. New in this round, the 43a1 (1,0)-type curve ($\mathbb{Q}$-side rank 1, E^5-side rank 0) admits a fully closed Lean proof: L_P(43a1,1) > 0 (genuine theorem LPrime43), $L(E^5,1) > 0$ and $L(E^5,1) \neq 0$ (LSeries5_43 fully closed), and the closure audit of the assembly L_prime_43_ne_zero contains neither refined-BSD-formula axioms nor GRH. The (1,0) type is, in the rank-symmetry framework, the intermediate instance of "rank additivity + one-sided rank 1": the K-side rank 1 is combined from the $\mathbb{Q}$-side rank 1 and the E^5-side rank 0. All the closed proofs above depend on no conjecture; the output files and recomputation scripts are preserved and fully reproducible.

4.2 Formalization skeleton system

Channel A (the BSD-formula path) and Channel B (the unconditional sign-preservation path) merge in the module PStarOr into a single theorem: if the formula holds or the unconditional structure holds, then P* holds. Main.lean closes the argument with a six-ring documentation chain: from the rank-two exclusion to the closure of P*, the two channels, the merge, extended sign preservation, and the retirement of GRH; each ring has its corresponding module and compilation record. The new assembly modules put rank symmetry into Lean: in SymmetryReduction, symbol_symmetry_11 is the symmetric restatement of the second-order Leibniz (1,1)-type cancellation (its closure audit contains no formula-conjecture axiom in practice), and the componentwise sign inputs of p_star_11_sym are mathematically already Kolyvagin's theorem on the rank-one side; SymmetryClosure assembles the three-piece set for the instance 79a1 (entanglement_79, Lpp_EK_79_pos, Lpp_EK_79_ne_zero, rank_EK_79_two, sha_finite_EK_79, symmetry_closure_79), whose closure contains only 79-series object-layer axioms (the datum zeroOrder plus theorem-level registrations leibniz_11_79/rank_additivity_79/Kolyvagin ×2/descent), with no refined-BSD-formula axiom and no GRH; SelmerDescent registers the exact sequences $0 \to Sel(E) \to Sel(E/K) \to Sel(E^5)$ and the descent sequence (Sel finite $\Longrightarrow$ Sha finite) at theorem level (same layer as Kolyvagin) and assembles sha_finite_EK_11_via_selmer; LeibnizThird supplies the

(2,1)/(1,2)-type third-order Leibniz genuine theorems (see 4.9). All modules compile (exit 0), with closure-audit reports attached.

4.3 Theoretical hypothesis reduction: the retirement of GRH

The key theoretical progress of our work is the retirement of GRH from the hypothesis family of P*. Core fact: the interval (1, 3/2) is contained in Re(s) > 1, and Re(s) > 1 is zero-free by a standard unconditional theorem of automorphic L-theory — if $L(s_0) = 0$ with $\mathrm{Re}\, s_0 > 1$, then, since the Euler product converges absolutely in Re(s) > 1 [6], some local factor $1 - \alpha_v \cdot q^{-s_0} = 0$, hence $\alpha_v = q^{s_0}$ and $|\alpha_v| = q^{\sigma_0} > q$; this contradicts the Ramanujan bound in the elliptic curve case (the Hasse bound, proved by Deligne in the Weil conjectures [5]). The argument uses only modularity, absolute convergence and the Hasse bound — all unconditional theorems. Consequently the sign-preservation argument of Channel B no longer needs GRH; P* is doubly immune to the refined BSD formula and to GRH, and the finiteness of Sha disappears from the dependence of P* (P* needs only the sign and the rank, not the formula equality).

4.4 Two-sided closure

Algebraic side: R_2 is the determinant of the 2×2 height matrix, and by positive definiteness of the height pairing $R_2 > 0$ (theorem level, geometric rigidity). Analytic side: $L''(E/K, 1) > 0$ (closed by instance proofs; the theoretical skeleton awaits a complete proof). The two-sided closure means: once a theoretical proof of $L''(1) > 0$ is completed, positivity originates from "rank exactly 2" rather than from a regulator collapse — this is a key step of analytic–algebraic consistency in the rank-two case of weak BSD, and the criterion by which we separate the two cases "rank-2 positivity" and "regulator degeneration". For the rank-two reverse direction, P* is precisely the obstacle-elimination tool of the reverse proof at the exclusion layer: if $L'' > 0$ is proved, the rank-two reverse direction lacks only the number-field version of the vanishing layer over K — namely that $L(E/K, \cdot)$ vanishes directly over K, which requires the Lean implementation of the number-field root number $\varepsilon_K = -1$, of the functional equation of L(E/K) and of the Artin decomposition, rather than merely the $\mathbb{Q}$-side vanishing through the product $L(E,1) \cdot L(E^5,1) = 0$.

4.5 Half-by-half comparison of the rank-2 positivity expansion, and route assessment

Whether rank 2 can be expanded like rank 1 requires a half-by-half comparison. The complete rank-1 positivity chain is: non-vanishing ($L'(1) \neq 0$, the Euler system of Kolyvagin and Gross–Zagier, a deep theorem) times sign (the refined formula $L'(1) = \Omega \cdot R \cdot |Sha|/|T|^2$, whose right-hand side is entirely positive, the rank-one Sha finiteness being already Kolyvagin's theorem). The corresponding rank-2 chain is: non-vanishing ($L''(1) \neq 0$, Taylor structure, free at the level of definitions, cheaper than rank 1) times sign (the formula $L''(1)/2 = \Omega \cdot R_2 \cdot |Sha|/|T|^2$, whose right-hand side is entirely positive, except that the finiteness of Sha in the $\mathbb{Q}$-side rank-two component is completely open — apart from the K-side (1,1) decomposition, see 4.7). Hence the only point where rank 2 lags behind in full generality is the finiteness of Sha in the $\mathbb{Q}$-side rank-two component — precisely the one genuine mathematical gap of P*. Currently no general (non-CM) elliptic curve of rank ≥ 2 has an unconditional proof of the finiteness of Sha; the CM case (Rubin's Euler systems and the Iwasawa main conjecture) reaches only rank ≤ 1.

A supplement (project verification of 2026-10-11): for the $\mathbb{Q}$-side rank-two component (the 389a1 isogeny class), the project has introduced at the document/discussion level an A_5 non-abelian Artin representation framework to optimize its sign and parity structure. The A_5-type 2-dimensional irreducible Artin representations ($GL(2,\mathbb{C})$-image the binary icosahedral group $2.A_5 \cong SL(2,5)$, projective image $A_5 \cong PSL(2,5)$; A_5 itself has no 2-dimensional complex irreducible representation, and its irreducible complex representations have dimensions exactly 1, 3, 3′, 4, 5, with trace field $\mathbb{Q}(\sqrt{5})$) are automorphic by the theorem of Deligne–Serre (weight-1 newforms, theorem level), so a non-abelian rank-allocation tool exists; the non-abelian formula (4) reduces the parity of the K-side rank to ord L(E,1) + ord L(E,3) + ord L(E,3′) + ord L(E,5) (mod 2), and the 4-dimensional component ($= \rho_1 \otimes \rho_2$) has root number identically +1, hence its analytic rank is necessarily even; for $N_E = 389$ (prime), the negative root number comes uniquely from the local correction factor at the ramified prime 389 — the "unique true gap at the bad prime 389" of 389a1 is thereby structured by the A_5 decomposition; DR2 Theorem 6.11 further provides an unconditional non-abelian rank-preservation baseline (equal analytic ranks $\Longrightarrow$ equal arithmetic ranks). The framework optimizes the sign and parity layer; it does not change the fact that the finiteness of Sha for rank ≥ 2 remains open — the latter remains the one genuine mathematical gap of P*.

The “Heegner point of rank 2” has only the following promising routes, all conditional or open. First, triple/diagonal cycles (Gross–Schoen; Yuan–Zhang–Zhang): the diagonal class in $CH^2(X\times X\times X)$ gives a “rank-2 Heegner class”, and the diagonal-cycle formula, under the hypothesis that the diagonal period is non-zero, implies rank = 2 and the finiteness of Sha — a verbatim analogue of the rank-1 expansion, but the non-vanishing hypothesis itself has no unconditional proof. Second, p-adic/Iwasawa main conjectures: controlling Sha[p^∞] prime by prime via the p-adic Gross–Zagier analogue depends on the main conjecture and $\mu = 0$-type hypotheses, and is open for every p for general rank-≥ 2 curves. Third, computational descent: it supplies only Sha[n] for finitely many n and can never reach the overall finiteness. Conclusion: treating the finiteness of Sha in the $\mathbb{Q}$-side rank-two component as a prerequisite of P* would drag the whole project into the very center of the BSD rank-two frontier of number theory.

Our strategic judgment is therefore: P* needs the sign, not the finiteness of Sha. Channel B (the zero-free region $\mathrm{Re}(s) > 1$ plus IVT sign preservation) already provides a sign source that does not pass through the formula, and its Lean closure contains no refined-BSD-formula axiom in practice — the finiteness of Sha enters P* only through the formula path of Channel A, and only in the form “finite $\Longrightarrow$ well-defined positive real number”, without requiring the order of Sha. For P*, the remaining feasible work is to realize the theorem-level registrations of Channel B (the zero-free region, the Kolyvagin rank-one non-vanishing, the analytic object layer), not to prove the finiteness of Sha in the $\mathbb{Q}$-side rank-two component; if the goal is later changed to positivity for a specific rank-2 curve (e.g. $L''(1) \approx 1.5186 > 0$ for 389a1), the rigorous interval-arithmetic verification path applies, and again the finiteness of Sha does not enter the dependence.

4.6 Corollary A and Corollary C

Corollary C (sign preservation under extension by the functional equation): the real-coefficient functional equation $\Lambda(\sigma) = \varepsilon\cdot\Lambda(2-\sigma)$ extends the right-half zero-free region to $(1/2, 3/2)\setminus\{1\}$: the sign and the zeros of the left half are pulled back from the right half through the root number ε; its formalization is complete and has passed the closure audit. Corollary A (lower bound on zeros): the Hadamard logarithmic identity combined with the positivity margin of the components gives the lower bound $\gamma_0 \geq \delta_0/\sqrt{(\exp(2(U+E))-1)}$ for the height of the nearest non-central zero, where δ_0 is a chosen real-axis shift and U, E numerical upper bounds; its Lean skeleton (module ZeroGap) is on disk, and the closure audit confirms that it

contains only object-layer and Hadamard axioms — no formula conjecture and no GRH, fully independent of the P* hypothesis chain.

4.7 Twist-involution rank symmetry: rank 2 assembled from rank 1

The Galois characters of a quadratic extension decompose the K-side rank-2 problem symmetrically into two $\mathbb{Q}$-side rank-1 problems. Let χ be the quadratic character of $K/\mathbb{Q}$ and let the twist operation σ: $E \mapsto E^5$ (the χ-twist, an involution with $(E^5)^5 = E$). Then $E(K)$ as a $\mathbb{Z}[\mathrm{Gal}(K/\mathbb{Q})]$-module decomposes by characters into $E(\mathbb{Q}) \oplus E^5(\mathbb{Q})$, which gives the rank additivity symmetry: rank $E(K)$ = rank $E(\mathbb{Q})$ + rank $E^5(\mathbb{Q})$ (Galois character decomposition, a theorem); on the analytic side the Artin decomposition gives ord $L(E/K,1)$ = ord $L(E,1)$ + ord $L(E^5,1)$ (in Lean the axiom zeroOrder_additivity). The K-side rank 2 distributes over the $\mathbb{Q}$ side in three types: (2,0), (1,1), (0,2).

The (1,1) type (both E and E^5 of analytic rank 1) is the most valuable branch of the rank symmetry; all three channels are assembled symmetrically from rank-one theorems. First, sign symmetry: from $L(E,1) = L(E^5,1) = 0$ the product second derivative degenerates to $L''(E/K,1) = 2 \cdot L'(E,1) \cdot L'(E^5,1)$ — P* in the (1,1) type is exactly $P_1^* \otimes P_1^*$, the entangling product of two rank-one positivities. The Lean version of this identity was completed with leibniz_11_second_object: the second-order Leibniz (1,1)-type cancellation is a pure-calculus genuine theorem (its symmetric restatement symbol_symmetry_11 has a closure containing no formula-conjecture axiom in practice), and the object-level assembly registers only the Artin-decomposition structural axiom. The sign inputs of the P* positivity branch (p_star_11_sym) still come from the component-level axioms of Positivity — their mathematical content on the rank-one side is already Kolyvagin's theorem (the registration is an implementation gap), and the unconditional sign preservation of Channel B can substitute for it.

Second, rank symmetry: the two sides with ord = 1 give, via Kolyvagin–Gross–Zagier, both sides with rank = 1, and rank additivity gives rank $E(K) = 2$ and ord $L(E/K) = 2$ — the forward rank-two weak BSD on the K-side (1,1) type is closed symmetrically from rank-one theorems. Third, finiteness symmetry: the finiteness of $Sha(E/\mathbb{Q})$ and $Sha(E^5/\mathbb{Q})$ on both sides (Kolyvagin), through the quadratic descent exact sequence $0 \to Sel(E,\mathbb{Q},p^\infty) \to Sel(E/K,p^\infty) \to Sel(E^5,\mathbb{Q},p^\infty) \to 0$ (with finite corrections at bad primes p), yields the finiteness of $Sha(E/K)$ — under the K-side (1,1) decomposition, the only mysterious factor of the refined BSD formula is assembled from rank-one theorems.

This corrects the scope of "the finiteness of Sha in rank 2 is completely open" in 4.5: the finiteness of Sha that is completely open is that in the $\mathbb{Q}$-side rank-two component (the 389a1 isogeny class) and in the (2,0)/(0,2) allocations (the latter contain a $\mathbb{Q}$-side rank-two component and fall back to the 389a1 class; their sign/parity structure is characterized by the A_5 non-abelian Artin representation framework, see 4.5); under the K-side (1,1) decomposition the finiteness of Sha(E/K) is a theorem. The reverse direction is equally reduced: rank E(K) = 2 $\Longrightarrow$ ord L(E/K) = 2 in the (1,1) type reduces to two $\mathbb{Q}$-side rank-one reverse directions (the target of P_1*) — the rank-two reverse direction becomes rank-one reverse ×2. The price of the symmetry is clear: it shifts the rank-2 difficulty to "at least one $\mathbb{Q}$-side rank-one component", and within the class of modular curves the sign and finiteness parts of P_1* are already theorems.

4.8 Arbitrary-rank symmetry: complete closure $\Leftrightarrow$ n ≤ m

Whether the twist-involution symmetry generalizes to arbitrary rank n depends on three layers. The structural layer (rank additivity rank E(K) = Σ_χ rank E^χ($\mathbb{Q}$), Artin decomposition ord L(E/K) = Σ_χ ord L(E^χ), Selmer descent along χ) is a theorem for arbitrary n and arbitrary extension degree m = [K:$\mathbb{Q}$]. The sign layer P*_n (analytic rank r $\Longrightarrow$ L^(r)(1) > 0) is unconditional for every r: Channel B directly gives L^(r)(1)/r! = $\lim_{s\to 1^+}$ L(s)/(s−1)^r ≥ 0, and ord = r gives L^(r)(1) ≠ 0, hence L^(r)(1) > 0 — no formula, no GRH, and no dependence on the symmetry (in the sign layer the symmetry is only the structural identity of the entangling product).

Complete closure (rank additivity + two-sided rank reverse by Kolyvagin–Gross–Zagier + two-sided finiteness of Sha by Kolyvagin, all assembled from rank-≤ 1 theorems) $\Leftrightarrow$ the existence of an all-ones allocation (1, …, 1) $\Leftrightarrow$ n ≤ m. Pigeonhole principle: if n > m, some component has rank ≥ 2 and falls back to the 389a1 class (whose sign/parity structure is characterized by the A_5 non-abelian Artin representation framework, see 4.5; the finiteness of Sha remains open). For a quadratic field m = 2: n = 1 and n = 2 of type (1,1) are closed (79a1 is on disk), and n ≥ 3 contains a rank-≥ 2 component — (1,1) is the only non-trivial fully closed rank for a quadratic field, and our symmetry chain reaches its limit there. Difficulty shift: the K-side rank-n problem is approximately the maximal component rank, and the (1,n−1) type lowers the order by one; the first skeleton for n = 3 is the third-order Leibniz cancellation of the next subsection.

4.9 The n = 3 skeleton: third-order Leibniz (2,1)/(1,2)-type genuine theorems

As the first skeleton of the n = 3 difficulty shift, we turn the (2,1)-type Leibniz leading-term formula into a genuine theorem (module LeibnizThird): the complete third-order formula $(f\cdot g)'''(1) = f'''(1)\cdot g(1) + 3\cdot f''(1)\cdot g'(1) + 3\cdot f'(1)\cdot g''(1) + f(1)\cdot g'''(1)$ (product_third_deriv, a pure-calculus genuine theorem); the (2,1)-type cancellation ($f(1) = f'(1) = 0$, $g(1) = 0$) gives $(fg)'''(1) = C(3,2)\cdot f''(1)\cdot g'(1) = 3\cdot f''(1)\cdot g'(1)$ (leibniz_21_third), and the (1,2)-type symmetry ($f(1) = 0$, $g(1) = g'(1) = 0$) gives $3\cdot f'(1)\cdot g''(1)$ (leibniz_12_third), together with the non-vanishing propagation _ne_zero. The closure audits of the five theorems depend only on [propext, Classical.choice, Quot.sound] — pure logical axioms, zero axioms, zero formula, zero GRH, zero object-layer registration; it is the first zero-registration module on the n = 3 side (compare p_star_11_sym, which contains bsd_formula_rank_one/one5). The proof route is a manual three-layer argument (v4.34 has no iteratedDeriv abstraction layer): open-neighbourhood C^3 stratification → pointwise identity ×3 inside v → event equivalence → the six-element combination of HasDerivAt. The two subsequent object-layer steps: the third-order lift of L_fun_structure ($L'''(E/K,1)$ plus the Artin (2,1)-allocation axiom) and the third-order object interface of Positivity (pushing the Channel B sign layer to rank 3).

5 Boundaries and limitations

The following must be stated honestly. First, a skeleton is not a complete proof: in Channel B the Kolyvagin rank-one non-vanishing, the zero-free region and the analytic objects are registered as axioms — mathematically they are theorems, but in Lean they are registered because mathlib lacks the corresponding theory; this is an implementation gap, not a mathematical one, and it can be progressively eliminated as the library develops. Second, P* itself remains a conjecture-level proposition, and its genuine mathematical gap converges precisely to the two rank-two refined-BSD-formula axioms (bsd_formula_rank_two, bsd_formula_rank_two5); the unresolved part is the finiteness of Sha in the $\mathbb{Q}$-side rank-two component (the 389a1 isogeny class) — the sign/parity structure of this component is characterized by the A_5 non-abelian Artin representation framework (see 4.5), which does not change the Sha-finiteness gap itself; on the rank-one side, the Sha-finiteness parts of the two axioms are already Kolyvagin's theorem and the registration is an implementation gap. The Selmer descent of task 3 (exact sequences $0 \to Sel(E) \to Sel(E/K) \to Sel(E^5)$ plus the descent sequence) is a standard theorem mathematically, registered at theorem level in Lean because

the algebraic-number-theory library is missing (same layer as Kolyvagin); the closure audit of its assembly sha_finite_EK_11_via_selmer contains only the SelmerDescent object-layer axioms and zeroOrder, with no formula and no GRH — this registration does not constitute a mathematical gap; the (2,1)/(1,2)-type third-order cancellations of LeibnizThird are pure-calculus genuine theorems with closure consisting of pure logical axioms only, completely zero-registration. Arbitrary-rank boundary: complete closure $\Leftrightarrow n \le m$; for a quadratic field $m = 2$ the chain is exhausted ((1,1) is the only non-trivial closed rank); $n \ge 3$ contains a rank-≥ 2 component, and the difficulty shifts to the maximal component rank, whose first skeleton (third-order Leibniz) is closed with zero registration. Third, the instance closed proofs cover only concrete curves and do not constitute a theoretical generalization; the numerical end of Corollary A (the concrete value of η) still awaits the fixed slot of the numerical pipeline; the instance closed proof for E^5-side rank-2 curves (type (0,2)) is not yet complete. Fourth, the version of the vanishing layer over the number field K ($L(E/K, \cdot)$ vanishing directly at $s = 1$) awaits the Lean implementation of the functional equation and the Artin decomposition. All the gaps above are explicitly recorded in our project records.

6 Significance and outlook

We demonstrate a feasible path for narrowing the hypothesis family of a “positivity proof”: from “formula conjecture + GRH” to “unconditional structure + numerical data”, with verifiable machine evidence at every step. Methodologically, the dual-track structure of rigorous interval arithmetic and a theorem-prover skeleton offers an auditable working style for the study of conjectures in arithmetic geometry: the numerical closed proof answers “whether the instance holds”, and the skeleton closure answers “what the argument depends on”. Outlook: first, to advance the numerical end of Corollary A and the extension of instances, giving the zero lower bound an explicit numerical value; second, to progressively realize the theorem-level registrations of Channel B, depending on the development of mathlib, so that the unconditional branch of P* becomes a complete formal proof; third, to track the frontier progress of the triple-cycle and p-adic routes in the rank-two case, reserving an access point for a possible future proof of the finiteness of Sha in the $\mathbb{Q}$-side rank-two component; fourth, to complete the version of the vanishing layer over the number field K. These works also contribute practical cases of arithmetic-geometry formalization to the Lean community, and serve as a concrete sample for research on AI-assisted formal proof.

References

[1] Birch, B. J., Swinnerton-Dyer, H. P. F. Notes on elliptic curves. II[J]. Journal für die reine und angewandte Mathematik, 1965, 218: 79-108.

[2] Gross, B. H., Zagier, D. B. Heegner points and derivatives of L-series[J]. Inventiones mathematicae, 1986, 84(2): 225-320.

[3] Kolyvagin, V. A. Finiteness of E(ℚ) and Sha(E, ℚ) for a subclass of Weil curves[J]. Izvestiya Akademii Nauk SSSR, Seriya Matematicheskaya, 1988, 52(3): 522-540; Mathematics of the USSR-Izvestiya, 1989, 32(3): 523-541.

[4] Wiles, A. Modular elliptic curves and Fermat's last theorem[J]. Annals of Mathematics, 1995, 141(3): 443-551; Breuil, C., Conrad, B., Diamond, F., Taylor, R. On the modularity of elliptic curves over ℚ: wild 3-adic exercises[J]. Journal of the American Mathematical Society, 2001, 14(4): 843-939.

[5] Deligne, P. La conjecture de Weil. I[J]. Publications Mathématiques de l'IHÉS, 1974, 43: 273-307.

[6] Godement, R., Jacquet, H. Zeta functions of simple algebras[M]. Lecture Notes in Mathematics 260. Berlin: Springer, 1972.

[7] Cremona, J. E. Algorithms for modular elliptic curves[M]. 2nd ed. Cambridge: Cambridge University Press, 1997.

[8] The L-functions and Modular Forms Database (LMFDB)[EB/OL]. https://www.lmfdb.org.

[9] Birch and Swinnerton-Dyer Conjecture[EB/OL]. Clay Mathematics Institute. https://www.claymath.org/millennium/birch-and-swinnerton-dyer-conjecture/.

[10] Deligne, P., Serre, J.-P. Formes modulaires de poids 1[J]. Annales scientifiques de l'École Normale Supérieure, 1974, 7(4): 507-530.

[11] Darmon, H., Rotger, V. Diagonal cycles and Euler systems II[J]. Journal of the American Mathematical Society, 2017, 30(3): 601-672.